

Amirhossein Bagheri

M.SC. STUDENT IN COMPUTER SCIENCE & ENGINEERING

Milan, Italy

✉ amirhossein.bagheri.001@gmail.com | 🏠 ahabagheri01.github.io | 📄 ahabagheri01 | 🎓 Google Scholar

Education

Politecnico di Milano

M.Sc. IN COMPUTER SCIENCE & ENGINEERING

Milan, Italy

Sept. 2025 - Present

- GPA: 29.25/30

Sharif University of Technology

B.Sc. IN COMPUTER ENGINEERING

Tehran, Iran

Sept. 2019 - July 2024

- GPA: 19.58/20 (29.37/30); major GPA: 19.82/20 (29.73/30).
- Thesis: Enhancing lesion detection (segmentation) in breast mammograms through semi-supervised learning.

Research Interests

Interests Machine Learning Theory, Optimization, High-Dimensional Statistics, Reinforcement Learning

Activities

Polimi Data Scientists, Data Science Club

EVENT TEAM MEMBER

Politecnico di Milano

2025 - Present

- Part of the student-led data science association at Politecnico di Milano.

Publications

f -FUM: Federated Unlearning via Min-Max and f -Divergence

AMIRHOSSEIN BAGHERI, RADMEHR KARIMIAN, MEGHDAD KURMANJI, NICHOLAS D. LANE, AND GHOLAMALI AMINIAN

Under Review

2026

Enhanced Classifier-Guided Diffusion through Information-Theoretic Regularization

SEYED ALIREZA JAVID, AMIRHOSSEIN BAGHERI, AND NURIA GONZALEZ-PRELCIC

IEEE ISIT

2026

Enhancing Diffusion Model Guidance through Calibration and Regularization

SEYED ALIREZA JAVID, AMIRHOSSEIN BAGHERI, AND NURIA GONZALEZ-PRELCIC

NeurIPS SPIGM Workshop

2025

f -SCRUB: Unbounded Machine Unlearning via f -Divergences

AMIRHOSSEIN BAGHERI, RADMEHR KARIMIAN, AND GHOLAMALI AMINIAN

ICLR DATA-FM Workshop

2025

Semi-supervised Learning under Self-training via f -Divergence

GHOLAMALI AMINIAN, AMIRHOSSEIN BAGHERI, RADMEHR KARIMIAN, MAHYAR JAFARI NODEH, MOHAMMAD HOSSEIN YASSAEI

Tiny Papers Track at ICLR

2024

Robust Semi-Supervised Learning via f -Divergence and α -Renyi Divergence

GHOLAMALI AMINIAN, AMIRHOSSEIN BAGHERI, MAHYAR JAFARI NODEH, RADMEHR KARIMIAN, AND MOHAMMAD-HOSSEIN YASSAEI

IEEE ISIT

2024

Research Experience

AIRIC, Politecnico di Milano

FOUNDATION MODELS FOR TIME SERIES

Milan, Italy

Ongoing

- Benchmarking foundation models against classical statistical methods for time-series forecasting.
- Studying and comparing the architectures of different foundation models for time series.

Politecnico di Milano

ONLINE LEARNING IN SERVERLESS COMPUTING

Milan, Italy

Dec. 2025 - May 2026

- Involved in ongoing research on online learning methods for serverless computing systems.

The Alan Turing Institute

f -FUM: FEDERATED UNLEARNING VIA MIN-MAX AND f -DIVERGENCE

Remote

Feb. 2025 - May 2026

- Developed a federated unlearning framework formulated as a min-max optimization problem with f -divergence objectives.
- Evaluated the method across federated setups, showing speedups over retraining with minimal utility degradation.
- Paper currently under review at the NeurIPS main conference.

The Alan Turing Institute

f -SCRUB

Remote

Nov. 2024 - Feb. 2025

- Introduced a novel framework based on f -divergences with the SCRUB framework for machine unlearning.
- Conducted comprehensive experiments investigating different combinations of f -divergences in f -SCRUB.

JULY 16, 2026

AMIRHOSSEIN BAGHERI · CURRICULUM VITAE

1

- Proposed novel risk functions inspired by f -divergences and α -Renyi divergence.
- Provided an upper bound on ideal performance for metric distance.
- Proposed novel regularization terms inspired by f -divergences and α -Renyi divergence.
- Analyzed the empirical risk functions and regularizers across different scenarios and datasets under noisy pseudo-labels.

Sharif University of Technology

B.Sc. THESIS

Tehran, Iran
Jan. 2023 - May 2024; Defense: June 2024

- Thesis: Enhancing lesion detection in breast mammograms through semi-supervised learning.
- Developed a weakly supervised labeling pipeline to generate segmentation masks for unlabeled mammogram data.
- Introduced and evaluated a pipeline for using unlabeled data in lesion detection in mammograms.

Achievements

Sept. 2025 **MAECI Scholarship**, Ministry of Foreign Affairs of the Italian Government

Sept. 2024 **Ranked 5th**, Among 190 undergraduate students, Sharif University of Technology

June 2019 **Ranked 31st**, Nationwide Universities Entrance Exam (Konkur), among 160,000 participants

Technical Skills

Programming	Python, C++, C, Matlab, PostgreSQL, LaTeX, R, Java
Frameworks	LangChain, TGI, vLLM, Llama.cpp, SQLAlchemy, Neo4j, Git
DevOps & OS	Linux, Docker
Machine Learning	NumPy, Pandas, Scikit-learn, Matplotlib, MLflow, Seaborn
Deep Learning	PyTorch, Keras, TensorFlow
Languages	Persian (native), English (proficient; TOEFL 105: R28, L28, S27, W21)

Teaching Experience

Sharif University of Technology

TEACHING ASSISTANT AND PROJECT SUPERVISOR

Tehran, Iran
Spring 2022 - Fall 2023

- Differential Privacy (graduate): project supervisor, Fall 2023.
- Signals and Systems: head TA, Fall 2022 and Fall 2023.
- Probability and Statistics: designed and graded assignments, Fall 2022 and Fall 2023.
- Linear Algebra: project supervisor; designed and graded assignments, Spring 2022 and Fall 2023.
- Artificial Intelligence and Design Algorithm: designed and graded assignments; held TA sessions, Spring/Fall 2022.

Selected Coursework

PhD	Reinforcement Learning (28/30)
Master	Bayesian Statistics (29/30), Machine Learning (19.3/20), High-Dimensional Probability (18.6/20), Information-Theoretic Methods in Learning and Statistics (18.4/20), Stochastic Process (19/20)
Bachelor	Engineering Probability and Statistics (20/20), Linear Algebra (20/20), Discrete Structure (19.6/20), Artificial Intelligence (20/20), Design Algorithm (20/20), Advanced Information Retrieval (20/20), Introduction to Bio-Informatics (20/20)

Related Course Projects

Bayesian Statistics (Fall 2025): Bayesian modeling of transconjugation efficacy under chemical stressors.

High Dimensional Analysis (Spring 2023): Theoretical analysis of adversarial learning.

Information Theory, Statistics, Learning (Fall 2022): Application of tilted losses and α -Renyi divergence in robust learning.

Machine Learning (Fall 2021): Dimension reduction, SVMs, transfer learning, and recommendation systems.

Artificial Intelligence (Spring 2021): A*, genetic algorithms, CSP, minimax, Bayesian inference, Gibbs sampling, DNN, HMM, and RL.

Advanced Information Retrieval (Spring 2022): Classification, clustering, ranking, and advanced search of 2k articles with BERT.

Introduction to Bio-Informatics (Fall 2022): Microarray data analysis.

Data and Network Security (Spring 2023): Simplified implementation of Signal Messenger.

Compiler Design (Fall 2021): C-minus compiler.

Computer Network (Spring 2022): Streaming application using socket programming.

Operating Systems (Spring 2022): Semaphore, scheduler, and file-system components in Pintos.